

Full 5-torsion of the regular-pentagon quintic pencil

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Abstract

We compute all twenty-five geometric fifth-torsion points of the genus-one normalization of the regular-pentagon quintic pencil asked about in McCallum's Question 17 in the AIM workshop problem list on rational and integral points. For a fixed equation normalization over $K = \mathbb{Q}(\sqrt{5})$, we give a cubic model, both birational maps, a degree-ten polynomial and both ordinate branches for the twenty points outside the known infinity subgroup. The complete division field is $K(\sqrt{5 + 2\sqrt{5}}, \zeta_5, \sqrt[5]{\lambda + \phi^5})$, where $\phi = (1 + \sqrt{5})/2$. Its degree is four or twenty, with the exact splitting criterion and Galois action specified. The formulas include every elliptic normalization, including the allowed cuspidal plane member. The method uses classical Tate normal form, division-polynomial theory, Fisher's full-level cover and Verdure's torsion criterion. A reproducible verification supplement accompanies this unrefereed, extensively AI-assisted note.

1 The pencil and the answer

The original question concerns the normalization of a pencil formed by the five side lines of a regular pentagon and the square of its circumcircle equation [1, Question 17, p. 51]. The source already observes the five-point infinity subgroup. We compute the other twenty points and the extension data as well. We fix coordinates, scaling and origin explicitly; changing the scaling changes the numerical pencil parameter. We make no first-priority claim. The universal modular and division-polynomial ingredients below are classical. Verdure gives the universal full-torsion Kummer criterion, including specialization [3, Theorem 5, pp. 84–88]. Morton also gives explicit coordinates for all twenty nonmarked points on the Tate family [4, Theorem 2.1]; its residual polynomial is exactly the one below after a sign convention change. Our computation concerns the explicit bridge from this pentagonal plane pencil to that classical family and the resulting complete geometric and arithmetic answer.

Put

$$r = \sqrt{5}, \quad K = \mathbb{Q}(r), \quad \phi = (1 + r)/2, \quad c = \phi^5 = (11 + 5r)/2, \quad d = 5 + 2r, \quad \delta^2 = d.$$

Let $Q = X^2 + Y^2 - T^2$ and

$$P = 2(X^5 - 10X^3Y^2 + 5XY^4) + 5\phi(X^2 + Y^2)^2T - 5\phi^3(X^2 + Y^2)T^3 + cT^5. \quad (1)$$

The projective pencil is $\Gamma_\lambda : P + \lambda TQ^2 = 0$. The factor T is the homogenization of the affine degree-four term. Let E_λ denote its smooth projective normalization, with origin $O = [0 : 1 : 0]$.

Theorem 1. *In characteristic zero the finite members have a geometrically integral genus-one normalization exactly when*

$$\lambda \notin \{0, -5r, -c\}. \quad (2)$$

For every such member, equations (13) and (16)–(17), together with the four points (15) and the origin, compute all 25 geometric fifth-torsion points. For $\lambda \in K$ satisfying (2), choose a primitive fifth root ζ and θ with $\theta^5 = \lambda + c$. Then

$$K(E_\lambda[5]) = K(\delta, \zeta, \theta). \quad (3)$$

This has degree 4 when $\lambda + c \in K^{\times 5}$ and degree 20 otherwise. The respective Galois groups are $C_2 \times C_2$ and $D_{10} \times C_2$, where D_{10} has order ten. Moreover

$$E_\lambda(K)[5] = 0, \quad E_\lambda(K(\delta))[5] \cong \mathbb{Z}/5\mathbb{Z}, \quad (4)$$

the latter subgroup being the five infinity points. Generically, (3) holds over $K(\lambda)$ and has degree twenty.

The question's remarks about twists, local conditions and nonregular or star polygons are separate variants. We do not assert a Tate–Shafarevich construction or solve those variants. This specified model has the rational origin O .

2 Normalization and the exact elliptic range

To check that (1) is the side product, set $Z = X + iY$, $W = X - iY$ and choose $\zeta = e^{2\pi i/5}$. The factors $Z + \zeta^{2j+1}W - (1 + \zeta)\zeta^j T$, for $0 \leq j < 5$, vanish on successive unit-circle vertices. Their product is

$$Z^5 + W^5 + 5\phi Z^2 W^2 T - 5\phi^3 Z W T^3 + cT^5 = P.$$

Thus this is the required regular pentagon, with a fixed nonzero scaling.

In the chart $T = 1$, write the equation as $AY^4 + BY^2 + C = 0$, where

$$\begin{aligned} A &= 10X + 5\phi + \lambda, \\ B &= -20X^3 + 2(5\phi + \lambda)X^2 - 5\phi^3 - 2\lambda, \\ C &= 2X^5 + (5\phi + \lambda)X^4 - (5\phi^3 + 2\lambda)X^2 + c + \lambda. \end{aligned}$$

With $h = 4X^2 + 2X - 1$, $a = 40 + 20r + 8\lambda$, $b = a + 5$, direct expansion gives

$$B^2 - 4AC = h^2(20X^2 - aX + b), \quad V = \frac{2AY^2 + B}{h}, \quad t = V - 2rX, \quad X = \frac{b - t^2}{a + 4rt}. \quad (5)$$

Put $\alpha = 1 - r/5$, $\gamma = 2r/5$, $m = 5 - 2r$, $z = t + d$, $s = z + \alpha\lambda$, and $w = 20Y(z + \gamma\lambda)/z$. Define

$$\begin{aligned} F(z) &= z^3 - ((3 + r)\lambda + 40 + 20r)z^2 \\ &\quad + ((60 + 36r)\lambda + 900 + 400r)z + (48 + 16r)\lambda^2 + (500 + 220r)\lambda. \end{aligned}$$

Substitution in (5) gives

$$Y^2 = \frac{mz^2 F(z)}{400(z + \gamma\lambda)^2(z + \alpha\lambda)}, \quad w^2 = \frac{mF(s - \alpha\lambda)}{s}. \quad (6)$$

Set

$$\begin{aligned} a_0 &= 5 - 2r, & a_1 &= (-26 + 10r)\lambda - 20r, \\ a_2 &= (42 - 86r/5)\lambda^2 + (-100 + 100r)\lambda + 500 + 200r, \\ a_3 &= (-112 + 48r)\lambda^2(\lambda + 5r)/5. \end{aligned} \quad (7)$$

The coordinates $\xi = a_3/s$, $\eta = \xi w$ yield the monic cubic

$$W_\lambda : \quad \eta^2 = \xi^3 + a_2\xi^2 + a_1a_3\xi + a_0a_3^2. \quad (8)$$

The inverse map is

$$s = a_3/\xi, \quad z = s - \alpha\lambda, \quad t = z - d, \quad X = \frac{b - t^2}{a + 4rt}, \quad Y = \frac{\eta z}{20\xi(z + \gamma\lambda)}. \quad (9)$$

These are inverse function-field maps. For example, with $G = AY^4 + BY^2 + C$ and the forward value of t , the exact identity

$$b - t^2 - X(a + 4rt) = -4AG/h^2$$

proves the recovered X is unchanged; Y cancels literally. Conversely (8) recovers the conic branch in (6) and hence the same t, ξ, η . These fractions specify dense open charts; at poles their unique extensions between smooth projective normalizations are understood.

Here are the identities establishing the global, rather than selected component, normalization:

$$\begin{aligned} a^2 - 80b &= 64\lambda(\lambda + 5r), \\ F(-\alpha\lambda) &= (-16/5 + 16r/25)\lambda^2(\lambda + 5r), \\ \text{disc}(F) &= (24064 + 10752r)\lambda(\lambda + 5r)^3(\lambda + c), \\ \Delta_W &= 2^{24}(161 - 72r)\lambda^5(\lambda + 5r)^5(\lambda + c). \end{aligned} \quad (10)$$

Over an algebraic closure, the conic square root in (5) is nontrivial for $\lambda(\lambda + 5r) \neq 0$. Its field is rational in t . Under (2), the remaining square root in (6) has precisely four odd branch points: the three distinct roots of F and $s = 0$. Thus the original quartic in Y is irreducible over the algebraic closure of K , and Riemann–Hurwitz gives genus one. This also proves that (8) normalizes the whole plane curve.

At $\lambda = 0$ the curve is five side lines. At $\lambda = -5r$, (1) is the same line-product expression with ϕ replaced by $(1 - r)/2$, again five lines. At $\lambda = -c$ the conic is nonsingular and F has exactly one double root and no triple root; the gcd in the s -coordinate is $s - 1 - 3r/5$. Removing its square leaves two odd branch points, giving genus zero. The infinite pencil member is $TQ^2 = 0$. By contrast, at $\lambda = -(25 + 10r)/4$ the five plane vertex nodes become cusps but (10) is nonzero, so this member is included in the elliptic range.

At infinity on (8), $\text{ord}_O(\xi) = -2$, $\text{ord}_O(\eta) = -3$, so s vanishes to order two and w has a simple pole. In (9), $z = -\alpha\lambda$, $z + \gamma\lambda = (\gamma - \alpha)\lambda$ and $a + 4rt = 4(3 - r)\lambda$ are nonzero. Therefore Y has a simple pole and $X = -(5\phi + \lambda)/10$ is finite: the image is $[0 : 1 : 0]$. Its plane X partial is ten, so it is smooth. Thus the origins agree.

3 The actual Tate twist and all the torsion points

Define

$$\beta = \frac{(11 - 5r)\lambda}{2(\lambda + 5r)}, \quad k = \frac{4(\lambda + 5r)}{r}, \quad q = dk^2. \quad (11)$$

The Tate equation [2, Lemma 1.1 and (2), p. 172] is

$$D_\beta : \quad y^2 + (1 - \beta)xy - \beta y = x^3 - \beta x^2, \quad \Delta_\beta = \beta^5(\beta^2 - 11\beta - 1). \quad (12)$$

The following is an origin-preserving equation isomorphism over $K(\delta)$, valid on every allowed fiber:

$$\xi = q(x - \beta), \quad \eta = (k\delta)^3 \left(y + \frac{(1 - \beta)x - \beta}{2} \right). \quad (13)$$

Indeed, put

$$T_\beta(x) = 4x^3 + (\beta^2 - 6\beta + 1)x^2 + 2(\beta^2 - \beta)x + \beta^2. \quad (14)$$

The right-hand side of (8) at $\xi = q(x - \beta)$ is $q^3 T_\beta(x)/4$, and $(k\delta)^6 = q^3$. This verifies the quadratic twist, including its sign, rather than just a j -invariant equality. No extra exclusion at $j = 0$ or 1728 occurs.

The known nonzero points on D_β are

$$(0, 0), \quad (0, \beta), \quad (\beta, 0), \quad (\beta, \beta^2). \quad (15)$$

For $P_0 = (0, 0)$ the tangent is horizontal; the chord law gives $2P_0 = (\beta, \beta^2)$ and $3P_0 = (\beta, 0) = -2P_0$, proving exact order five. The other points are its nonzero multiples.

The residual polynomial $R_\beta(x)$ is Morton's $D_5(x; b)$ [4, section 2, p. 5 of the arXiv version] under $b = -\beta$. Write $R_\beta(x) = \sum_{j=0}^{10} r_j x^j$ with the following coefficients:

j	r_j	
10	5	
9	$5(\beta^2 - 5\beta + 1)$	
8	$\beta^4 - 7\beta^3 + 44\beta^2 - 38\beta + 1$	
7	$\beta(\beta^4 + 3\beta^3 - 26\beta^2 + 127\beta - 9)$	
6	$\beta^2(\beta^4 + 3\beta^3 + 19\beta^2 - 248\beta + 36)$	
5	$\beta^3(\beta^4 + 3\beta^3 - 71\beta^2 + 322\beta - 84)$	
4	$\beta^4(\beta^4 - 12\beta^3 + 94\beta^2 - 293\beta + 126)$	
3	$5\beta^5(\beta^3 - 10\beta^2 + 36\beta - 25)$	
2	$5\beta^6(2\beta^2 - 13\beta + 16)$	
1	$10\beta^7(\beta - 3)$	
0	$5\beta^8$.	(16)

For every root x take both ordinates

$$y = \frac{-((1 - \beta)x - \beta) \pm \sqrt{T_\beta(x)}}{2}. \quad (17)$$

Then apply (13) and, if desired, (9).

Lemma 1. *The ten roots of R_β and the two choices in (17) are distinct on every allowed fiber and give exactly the twenty fifth-torsion points outside (15) and the origin.*

Proof. Put $b_2 = \beta^2 - 6\beta + 1$, $b_4 = \beta^2 - \beta$, $b_6 = \beta^2$, $b_8 = -\beta^3$. The classical division polynomials give

$$\begin{aligned} \psi_3 &= 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8, \\ H_6 &= 2x^6 + b_2x^5 + 5b_4x^4 + 10b_6x^3 + 10b_8x^2 + (b_2b_8 - b_4b_6)x + b_4b_8 - b_6^2, \\ \psi_5 &= T_\beta^2 H_6 - \psi_3^3 = x(x - \beta)R_\beta. \end{aligned}$$

These identities can also be derived directly from the chord law on $v^2 = f(x)$: writing $x_2 = x(2P)$, $x_3 = x(3P)$ gives $\psi_3 = 4f(x - x_2)$, $H_6 = 16f^2v(2P)/v(P)$ and $\psi_5 = 4f\psi_3^2(x_2 - x_3)$. The supplement performs that independent generic calculation and checks each coefficient in (16). The exact resultants and evaluations are

$$\text{Res}_x(\psi_5, T_\beta) = \Delta_\beta^6, \quad \text{Res}_x(\psi_5, \psi_3) = \Delta_\beta^8, \quad R_\beta(0) = 5\beta^8, \quad R_\beta(\beta) = 5\beta^{12}.$$

For a finite point with $\psi_5 = 0$ these nonzero resultants justify the chord denominators and imply $x(2P) = x(3P)$. Equality $2P = 3P$ would imply $P = O$, impossible, so $3P = -2P$ and $5P = O$. Conversely a nonzero fifth-torsion point has those denominators nonzero and satisfies this equation. In characteristic zero [5] is separable of degree 25 [5, sections 5.5–5.6]; hence its kernel has 25 distinct points. Negation pairs the 24 nonzero points into twelve distinct abscissae. Thus ψ_5 has twelve simple roots, and the resultant with T_β makes both ordinate signs distinct. Removing the two marked abscissae leaves exactly the asserted twenty points. $\square$

There is no additional inverse-chart exclusion for these twenty points. After (13), direct substitution gives

$$z + \gamma\lambda = (\gamma - \alpha)\lambda \frac{x}{x - \beta}, \quad a + 4rt = 4r(z + \gamma\lambda).$$

Thus the poles $\xi = 0$ and $z + \gamma\lambda = 0$ occur only at the two known abscissae, which are not roots of R_β . The five infinity points themselves have $X/Y = 0, \pm\sqrt{5+2r}, \pm\sqrt{5-2r}$. They are smooth, and rotation by 72 degrees preserves the pencil and cycles them. Its action on the normalization is an order-five translation, since the origin-fixing automorphism group in characteristic zero has no element of order five. They therefore form the order-five subgroup in (4), in agreement with the source's infinity observation.

4 The full division field and specialization

We specify the modular input; the infinity line or a semisimplification alone would not determine this field. Choose ζ with $1 + \zeta + \zeta^{-1} = \phi$. Over a field containing ζ , the Tate curve with marked P_0 has five choices of Q_0 satisfying $e_5(P_0, Q_0) = \zeta$, namely $Q_0 + jP_0$. These form the finite etale degree-five full-level cover of the marked Tate parameter line away from its cusps. An elliptic automorphism fixing a point of order at least four is trivial [2, Lemma 1.1]; thus the cover is fine. Fisher's full-level description [2, section 2 and Lemma 3.4, p. 194] gives its parameter map

$$\beta = \tau \frac{f(\tau)}{g(\tau)}, \quad f = \tau^4 + 3\tau^3 + 4\tau^2 + 2\tau + 1, \quad g = \tau^4 - 2\tau^3 + 4\tau^2 - 3\tau + 1.$$

With $\varepsilon(\tau) = (\phi\tau + 1)/(\tau - \phi)$ and $\iota(v) = (cv + 1)/(v - c)$, both transformations are involutions and the exact identities are

$$\tau f(\tau)/g(\tau) = \iota(\varepsilon(\tau)^5), \quad \iota(\beta) = -\frac{1}{\lambda + c}. \quad (18)$$

Consequently the full-level cover is $u^5 = -1/(\lambda + c)$, with $u = \varepsilon(\tau)$. This identifies every allowed specialization, not merely a generic extension: both covers are the normalization of the same function-field cover over the smooth cusp-complement base, and both are finite etale there. The fiber is exactly the choices of the complementary basis vector, including split fibers and the unmarked exceptional j values. No generic irreducibility assertion is imposed on every specialization.

Write $a = \lambda + c \neq 0$. If $\theta^5 = a$, then $u = -1/\theta$ satisfies $u^5 = -1/a$. Thus the two radical fields coincide. For a fixed identification of the Kummer group, the classes satisfy $[-1/a] = [a]^{-1}$, rather than being equal as nontrivial fixed classes; the inversion of generator or coordinate must be stated. Morton's coordinate has $u_M^5 = (2b + 11 + 5r)/(-2b - 11 + 5r)$; with $b = -\beta$ this is $c(\lambda + c)$, so $u_M = \phi\theta$ supplies the same radical [4, (2.1), p. 6 of the arXiv version]. This is older universal Tate-family machinery, not a new radical solver.

Verdure gives an earlier, direct specialization criterion [3, Proposition 3, Corollary 1 and Theorem 5]. For the Tate curve (12) over a characteristic-zero field containing ζ , its full fifth torsion is rational

exactly when $(\beta - c)/(\beta + c^{-1})$ is a fifth power. In our convention

$$\frac{\beta - c}{\beta + c^{-1}} = -c(\lambda + c) = (-\phi\theta)^5.$$

The statement and proof explicitly cover every nonsingular specialization. The full torsion field over a field containing ζ and a marked fifth-torsion point has degree one or five. After adjoining this fifth root all torsion is rational; when the root is not in the base field the criterion excludes degree one. Thus this older result also proves that the exact Tate torsion field is the indicated radical field, for split and nonsplit fibers.

Let $L = K(\delta, \zeta)$. The fine-cover interpretation and (13) give $L(E_\lambda[5]) = L(\theta)$ in both directions: a cover point supplies a basis and a basis determines its cover point. The image of P_0 has

$$\xi_{P_0} = -q\beta \in K, \quad \eta_{P_0} = -k^3 d\delta\beta/2.$$

The latter has a nonzero K coefficient, so the full torsion field contains δ . Nondegeneracy of the Weil pairing supplies ζ [6, Theorem 23.29]. Hence it already contains L , proving (3).

For completeness, $M = K(\delta)$ is quadratic because the norm of d to $\mathbb{Q}$ is five, not a rational square. It is real, while $K(\zeta)/K$ is imaginary quadratic; hence $[L : K] = 4$. If $v \in L$ and $v^5 = a \in K^*$, then $N_{L/K}(v)^5 = a^4$, so $(a/N_{L/K}(v))^5 = a$. Thus a becomes a fifth power in L exactly when it is one in K . Kummer theory now gives degrees 4 or 20. In the nonsplit case $K(\zeta, \theta)/K$ has group D_{10} : a rotation multiplies θ by ζ and the involution fixing the real fifth root inverts ζ . Its unique quadratic subfield is $K(\zeta)$, distinct from M , so adjoining δ gives $D_{10} \times C_2$. The same proof works generically; the valuation of $\lambda + c$ at $\lambda = -c$ is one, so it is not a fifth power in $K(\lambda)$.

The independent involution changing δ and fixing the Tate torsion field acts by $-I$ under (13). Therefore there is no nonzero K -rational fifth torsion. Over the real field M the known subgroup exists and $E(\mathbb{R})[5]$ has exactly five elements, so it is the entire M -rational subgroup. Finally, choosing the infinity generator as the first basis vector and adjusting the second to be anti-invariant under complex conjugation gives the matrices

$$\sigma_\delta \mapsto -I, \quad \sigma_\zeta \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_5 \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The nonzero transvection is normalized by choosing its generator. In the split degree-four case it is omitted. These matrices give the full extension, with the basis and generator choices made explicit.

5 Verification and research status

The accompanying package supplies exact symbolic geometry and generic chord-law checks, independent finite-field group-law enumerations, the arithmetic field controls, and deliberate mutants. Universal proofs establish the characteristic-zero all-fiber claims; finite tests are falsifiable implementation checks, not proofs by sampling. The arithmetic controls are also reproducible with the Python standard library. The package records dependencies, expected mathematical outputs and source references without redistributing third-party primary PDFs.

This note assembles an explicit computation for the source's regular-pentagon pencil, extending the already observed infinity subgroup to the complete kernel and division field. Tate normal form, Fisher's modular description, Verdure's full-torsion specialization criterion, Morton's explicit Tate-family formulas and residual polynomial, the division-polynomial theory and the Weil pairing are credited above. The bounded priority audit in the supplement documents its actual searches and reading limits; no absence-of-hits argument certifies first discovery or present-day openness. The repository's earlier partial infinity and quotient calculation is acknowledged as an input to the original derivation.

AI use and review. AI tools were used extensively in derivation, computation, literature research, writing, reproduction and adversarial review. The author assumes responsibility for the submitted claims and artifacts. This is an unrefereed preprint. Automated adversarial reviews do not constitute independent external human peer review, and no such human review or formal proof-assistant certification is claimed.

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